Leveraging Gaming to Enhance Knowledge Graphs for Explainable Generative AI Applications

Steph Buongiorno
Guildhall
Southern Methodist University
Dallas, United States
sbuongiorno@smu.edu

Corey Clark
Computer Science and Guildhall
Southern Methodist University
Dallas, United States
coreyc@smu.edu

***Abstract*—External knowledge graphs (KGs) can be used to augment large language models (LLMs), while simultaneously providing an explainable knowledge base of facts that can be inspected by a human. This approach may be particularly valuable in domains where explainability is critical, like human trafficking data analysis. However, creating KGs can pose challenges. KGs parsed from documents may comprise explicit connections (those directly stated by a document) but miss implicit connections (those obvious to a human although not directly stated). To address these challenges, this preliminary research introduces the GAME-KG framework, standing for "Gaming for Augmenting Metadata and Enhancing Knowledge Graphs." GAME-KG is a federated approach to modifying explicit as well as implicit connections in KGs by using crowdsourced feedback collected through video games. GAME-KG is shown through two demonstrations: a Unity test scenario from *Dark Shadows*, a video game that collects feedback on KGs parsed from US Department of Justice (DOJ) Press Releases on human trafficking, and a following experiment where OpenAI's GPT-4 is prompted to answer questions based on a modified and unmodified KG. Initial results suggest that GAME-KG can be an effective framework for enhancing KGs while simultaneously providing an explainable set of structured facts verified by humans.**

***Index Terms*—Human Computation Gaming, Crowdsourcing, Knowledge Graphs, Generative AI**

I. Introduction

Large language models (LLMs) have demonstrated remarkable capabilities in performing various natural language processing (NLP) tasks, including extracting semantic information and inferring cohesive text responses based on human input [11]. However, their application within domains where accuracy is critical–such as human trafficking data analysis–is limited due to a lack of explainability in results. This is because a LLM's mechanism for generating responses is opaque and may contain unwanted biases or contribute to "hallucinations" (generated text that is semantically possible but factually incorrect) [13]. High-stakes domain problems demand verifiable methods, making reliance on LLMs without transparency problematic. For instance, claims that someone broke a law could be harmful if they are based on hallucinations or lack validation. Explainability is crucial for trusting LLM generated results and validating them when needed [6], [7].

Providing LLMs with external, structured representations of facts in the form of knowledge graphs (KGs) has proven useful for addressing this limitation [8]. KGs are graphs of data that collect and convey knowledge. They represent information as structured relations between entities, which make a "domain model," along with schematic information, which make up an "ontology" [16]. Information can be retrieved from KGs to augment LLM generated responses in an explainable way [12], [19]. Human analysts can visually interpret the connections between nodes of the graph and view the structured path of facts from which information was derived [3].

While using KGs to augment LLM responses has proven beneficial, obtaining KGs that represent the desired domain information poses challenges. KGs can be parsed from documents, but capturing the many entity relationship from within text may not be feasible with computational approaches alone [22]. For instance, a KG parsed from text may include explicit relations between entities (e.g. semantic relations as directly expressed in words), but may not include implicit relations (e.g. relations that may be clear to a human, but not to the computer). In other cases it may be desirable to modify the explicit relationships of a KG, for instance, if the connections between entities are incorrect.

For these reasons, this research introduces the GAME-KG framework (standing for "Gaming for Augmenting Metadata and Enhancing Knowledge Graphs"). GAME-KG is a federated approach to modifying KGs that facilitates the collection of explicit and implicit knowledge. It leverages Human Computation Gaming (HCG)–a method of collecting feedback from crowds through video games–to modify and validate KGs [4]. GAME-KG guides the process of parsing a KG to presenting the data to the player and collecting feedback.

This research explores GAME-KG's potential across two demonstrations. The first demonstration presents a video game test scenario from the HCG, *Dark Shadows*, designed using the GAME-KG framework. *Dark Shadows* is a film noir-style mystery game that collects player feedback for modifying and validating KGs parsed from the US Department of Justice

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(DOJ) Press Releases on human trafficking. It features a drag-and-drop mechanic to demonstrate how video games can collect feedback and modify KGs. For the second demonstration, OpenAI's GPT-4 is asked questions related to the human trafficking press releases. GPT-4 is prompted to respond based on an unmodified KG parsed from text, and then a human-modified KG that encodes implicit relationships between entities. The initial results from these demonstrations suggest that a HCG designed via GAME-KG can be an effective way to collect human feedback, and that information retrieved from KGs can augment LLM responses in an explainable way, which is important for high-stakes domains like human trafficking data analysis.

II. Background

This section offers a brief overview of KGs and explainability, followed by a discussion on the value of crowdsourcing feedback for modifying and validating KGs.

A. Knowledge Graphs as Explainable Domain Models

A KG is a graph composed of nodes and edges, where nodes represent entities, and the edges between nodes convey the semantic relationship between entities. KGs model domains in the sense that their relations encode information on a subject. For example, a KG representing the domain of data science might include relationships between related scientific software packages, problem solving strategies, and metrics. Whereas, a KG for GIS might include relations between geographical entities, documents describing terrain, and coordinate systems.

Structured graphs have the added benefit of explainability. LLMs are complex, "black-box" systems and responses are generated without insight into the model's inner workings. Their high complexity makes model interpretation challenging [21]. Explainability is important, however, because it can make unintended biases evident, or it can reveal the information on which a response was based. Presented with a KG, human analysts can visually interpret the connections between nodes and read the structured path of facts from which information was derived [3]. This approach gives insight into the individual entity relationships that encode knowledge, and it supports the development of more reliable and trustworthy models for critical applications.

B. Gamification and Human Computation Gaming (HCG)

Obtaining KGs that represent explicit and implicit knowledge is not a trivial task. A KG parsed using computational methods alone may benefit from human feedback to connect or validate the relationships between nodes [22]. Crowdsourcing has been used to construct KGs manually, as in Wikidata [20]. However, motivating a crowd to provide feedback poses challenges. Gamification is an effective method for motivating crowds and collecting large amount of heterogeneous knowledge to achieve common goals [15]. Gamification has been widely used across diverse subjects–including education, healthcare, and marketing–to drive participation [5], [23].

HCG gamifies crowdsourcing tasks using video games. These games incorporate engaging gameplay mechanics that not only entertain players but also facilitate the collection of specific feedback required for various tasks. Adopting HCG can be advantageous because it leverages the vast global gaming community, which includes over three billion players, with predicted annual +3.7% player growth [14].

C. Crowdsourcing Implicit Knowledge

Crowdsourcing has proven to be successful for collect knowledge that may be intuitive to a human but not a machine [1]. Such intuitive knowledge may be especially useful for processing documents, as texts typically contain both explicit information (which is directly stated) and implicit information (which must be inferred but is not openly expressed) [18]. This dual layer of understanding helps in extracting deeper meaning and insight from documents.

Implicit information is derived between a claim and background knowledge. For instance, if a document claims "John Doe trafficked humans," it might not explicitly state that he broke the law. However, a human reader, possessing background knowledge, understands that human trafficking is illegal. This allows the reader to infer that John Doe's actions were unlawful. This implicit knowledge can then be incorporated into a knowledge graph (KG), enhancing its depth and utility by including implicit knowledge that is not directly mentioned in the text.

It is common that information is omitted from documents if it is assumed to be evident to the recipient, such that only parts of a message may be explicitly stated in words [1]. In result, a KG parsed from a document may not capture the many implied entity relationships expressed in a document. While LLMs have demonstrated the ability to perform inference and emulate common-sense reasoning responses, this ability is not exhaustive as a LLM may not possess the domain knowledge needed for certain implied information, indicating the need for crowdsourced human feedback [1].

III. The GAME-KG Framework

GAME-KG, shown by Figure 1, is a six-step federated approach to knowledge acquisition that leverages HCG to modify and validate KGs with the goal of enhancing them for high-stakes domain tasks in downstream generative AI applications. A HCG developed using the GAME-KG framework enables the collection of human feedback for modifying explicit and implicit entity relationships in a KG.

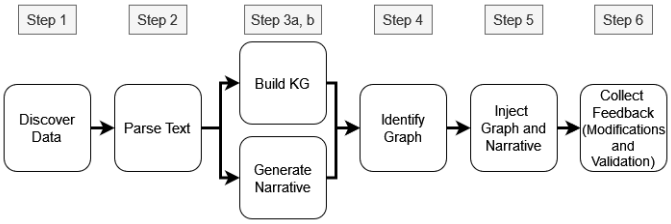

Fig. 1. The steps of the GAME-KG Framework.

The six-steps of GAME-KG are as follows:

- (Step 1: Discover Data) The document(s) that will be used to develop a domain-model are identified and collected.
- (Step 2: Parse Text) Entities and their relationships are parsed from the document(s) using am automatic, computational method [22].
- (Step 3a: Build KG) The parsed text is used to construct a KG. Entities are nodes and their relations are edges.
- (Step 3b: Generate Narrative) The parsed text is provided to an LLM instructed to generate a new, fictionalized narrative for the HCG. Research has shown that LLMs can generate stories for video games [2], [10]. Using a state-of-the-art LLM (e.g. GPT-4) and limiting narrative generation to small batches that fit within the LLM's context window can also mitigate against the generation of hallucinations inside the game narrative [9]. For GAME-KG, this step is used to generate narrative that is more palatable for gameplay while using the original entity relationships. For human inspection, the content can be anonymized and fictionalized within the game narrative.
- (Step 4: Identify Graph) The entities and relations making up the KG can be presented to the player for inspection. Specific graph sections may particularly benefit from human input. These sections can be identified based on a scoring method, such as cosine similarity. Low cosine similarity between connected nodes suggests incorrect connections, while high similarity between unconnected nodes suggests missing links. During gameplay, humans can judge whether a connection should be modified.
- (Step 5: Inject Graph and Narrative) The entities and relations as well as the fictional narrative are injected into the game.
- (Step 6: Collect Player Feedback (Modifications and Validation)) Each time a player suggests changes–based on explicit or implicit knowledge–these changes are saved as a weighted value. Connections between nodes with a low weight can be filtered out. This represents a federated approach where, based on consensus, the changes are retained or omitted.

IV. Demonstration I: GAME-KG Test Scenario

This section presents a test scenario from the HCG, *Dark Shadows*. *Dark Shadows* is a film noir-style mystery game that leverages GAME-KG to collect player feedback that modifies and validates KGs parsed from US DOJ Press Releases on human trafficking. In preparation of presenting narrative to the player, a subsection of the KGs are provided to GPT-4 with instruction to generate a fictional and anonymized narrative that retains the original entity relations. The resulting narrative contains key information meant for human inspection, but fictionalizes it for player engagement. As an example, the generated narrative might feature fictional events but contain real relationships between entities.

In *Dark Shadows*, the generated narrative is presented to the player as a fictional case briefing, as shown by Figure 2. Using a drag-and-drop mechanic resembling an inter-connected web, or "evidence board", players can select entities from the case file and specify connections between them by dropping them into the web. Weight can be added to the human-made connections as desired for validation, supporting the ability to collect graphs and also filter out responses with a low weight.

As a HCG, *Dark Shadows*'s design is based on the premise that gamification–in this case, progressing through a fictional narrative–can motivate players into providing feedback [23]. To foster this engagement, *Dark Shadows* also uses a generative, text-to-image model, Stable Diffusion, to generate in-game images matching the entities selected by the player [17].

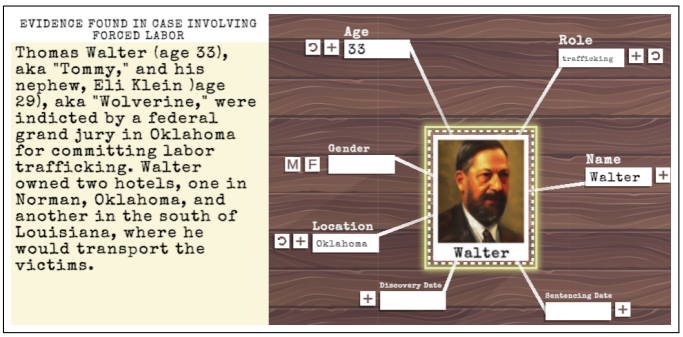

Fig. 2. Players are presented a fictional narrative that retains entity relations and can modify a KG using a web-like mechanic. The image of a "suspect" is generated with Stable Diffusion.

V. Demonstration II: Preliminary Results Evaluation through Q&A

The web-like mechanic can be used to modify explicit or implicit entity relations in a KG. This preliminary research focuses on modifying implicit entity relations and compares GPT-4's responses when augmented by information retrieved from a human modified graph and an unmodified KG. A KG automatically parsed from a US DOJ press release data serves as the example.

The data presents the claim that Kizer transported victims across state boarders, and that Villaman was an accomplice to Kizer. Kizer, the identified trafficker, is explicitly described as breaking the Mann Act for transporting a trafficked victim across state boarders. Villaman, another trafficker, is an accomplice to Kizer, but his relation to the Mann Act is not explicitly stated. Instead, the text encodes the implicit knowledge that if Villaman participated in human trafficking with Kizer, then Villaman also broke the Mann Act. To enhance the KG, a player connected Villaman with the Mann Act, as shown by Figure 3.

The modified KG was provided to GPT-4 for question-answering (Q&A). The information retrieved from the KG was used to augment generated responses as well as guard against hallucinations. When provided the human modified KG and asked which act Villaman broke, GPT-4 responded with the name of the act, "The Mann Act." When prompted to answer the same question based on the unmodified graph–in which Villaman is not connected to Mann Act–the LLM was unable to find the answer. Therefore, it was instructed to inform the

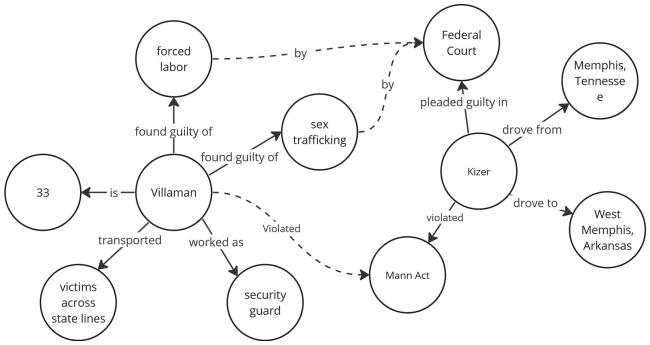

Fig. 3. An abstracted KG based on press release data. Solid lines represent explicit relationships, and dashed lines represent human modified connections.

user that the KG does not contain the knowledge to answer the question instead of hallucinating a response. This approach may be especially beneficial for high-stakes domains because structured representations of facts, like KGs, are explainable, can mitigate against hallucinations, and can encode human validation.

The validation methodology can vary based on research needs. The HCG method is suitable for a federated approach, where modifications to the graph are made if they receive enough votes from players. Additionally, these votes can be reviewed by an expert who can either confirm or reject the proposed changes.

VI. Discussion and Future Work

The GAME-KG framework serves as one step towards leveraging Human Computation Gaming (HCG) to modify KGs for accuracy critical generative AI applications, such as human trafficking data analysis. Obtaining KGs that have undergone human modification and validation may be especially desirable for high-stakes applications where accuracy is critical, as there is an emphasized need for explainablility. In the future, research will need to explore the additional benefits and risks of leveraging human feedback for modifying KGs. This research can be used to better understand situations where humans may contribute desirable biases, as well as undesirable biases. Taken together, the research presented here, as well as future work, can create a deeper foundational understanding of HCG leveraged for KG modification.

References

[1] Maria Becker, Katharina Korfhage, and Anette Frank. Implicit knowledge in argumentative texts: An annotated corpus. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 2316–2324, Marseille, France, 2020. European Language Resources Association.

[2] Steph Buongiorno, Lawrence (Jake) Klinkert, Zixin Zhuang, Tanishq Chawla, and Corey Clark. Pangea: Procedural artificial narrative using generative ai for turn-based video games. *Under Review*.

[3] Xu Chen, Hanxiong Chen, Hongteng Xu, Yongfeng Zhang, Yixin Cao, Zheng Qin, and Hongyuan Zha. Personalized fashion recommendation with visual explanations based on multimodal attention network: Towards visually explainable recommendation. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 765–774. ACM, 2019.

[4] Corey Clark, Ira Greenberg, and Myque Ouellette. A model for integrating human computing into commercial video games. In *2018 IEEE 6th International Conference on Serious Games and Applications for Health (SeGAH)*, pages 1–8, 2018.

[5] Hua Jonathan Ye Congcong Yang and Yuanyue Feng. Using gamification elements for competitive crowdsourcing: exploring the underlying mechanism. *Behaviour & Information Technology*, 40(9):837–854, 2021.

[6] Mathew P. Daggett and Frederick R. Waugh. The Human Trafficking Technology Roadmap: A Targeted Development Strategy for the Department of Homeland Security. Technical report, MIT Lincoln Lab, 2019.

[7] Amy Farrell and Rebecca Pfeffer. Policing human trafficking: Cultural blinders and organizational barriers. *The ANNALS of the American Academy of Political and Social Science*, 653(1):46–64, 2014.

[8] Yingqiang Ge, Wenyue Hua, Kai Mei, jianchao ji, Juntao Tan, Shuyuan Xu, Zelong Li, and Yongfeng Zhang. Openagi: When llm meets domain experts. In A. Oh, T. Neumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 5539–5568. Curran Associates, Inc., 2023.

[9] Anisha Gunjal, Jihan Yin, and Erhan Bas. Detecting and preventing hallucinations in large vision language models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(16):18135–18143, 2024.

[10] Vikram Kumaran, Dan Carpenter, Jonathan Rowe, Bradford Mott, and James Lester. End-to-end procedural level generation in educational games with natural language instruction. In *2023 IEEE Conference on Games (CoG)*. IEEE, 2023.

[11] Xiang Lorraine Li, Adhiguna Kuncoro, Jordan Hoffmann, Cyprien de Masson d'Autume, Phil Blunsom, and Aida Nematzadeh. A systematic investigation of commonsense knowledge in large language models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 11838–11855. Association for Computational Linguistics, December 2022.

[12] Robert Logan, Nelson F. Liu, Matthew E. Peters, Matt Gardner, and Sameer Singh. Barack's wife hillary: Using knowledge graphs for fact-aware language modeling. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5962–5971. Association for Computational Linguistics, July 2019.

[13] Marc Najork. Generative information retrieval. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '23, page 1, New York, NY, USA, 2023. Association for Computing Machinery.

[14] Newzoo. Global games market report. Market report, Newzoo, Jan 2024.

[15] Allard Oelen, Markus Stocker, and Sören Auer. Creating and validating a scholarly knowledge graph using natural language processing and microtask crowdsourcing. *International Journal on Digital Libraries*, 2023.

[16] Ciyuan Peng, Feng Xia, Mehdi Naseriparsa, and Francesco Osborne. Knowledge graphs: Opportunities and challenges. *Artificial Intelligence Review*, 56:13071–13102, 2023.

[17] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models, 2022.

[18] Keshav Singh, Farjana Sultana Mim, Naoya Inoue, Shoichi Naito, and Kentaro Inui. Exploring methodologies for collecting high-quality implicit reasoning in arguments. In *Proceedings of the 8th Workshop on Argument Mining*, pages 57–66. Association for Computational Linguistics, 2021.

[19] Fabian Suchanek and Anh Tuan Luu. Knowledge bases and language models: Complementing forces. In *Rules and Reasoning: 7th International Joint Conference, RuleML+RR 2023, Oslo, Norway, September 18–20, 2023, Proceedings*, pages 3–15. Springer, 2023.

[20] Denny Vrandečić and Markus Krötzsch. Wikidata. *Communications of the ACM*, 57(10):78–85, 2014.

[21] Haiyan Zhao, Hanjie Chen, Fan Yang, Ninghao Liu, Huiqi Deng, Hengyi Cai, Shuaiqiang Wang, Dawei Yin, and Mengnan Du. Explainability for large language models: A survey. *ACM Transactions on Intelligent Systems and Technology*, 15(2):1–38, 2024.

[22] Lingfeng Zhong, Jia Wu, Qian Li, Hao Peng, and Xindong Wu. A comprehensive survey on automatic knowledge graph construction. *ACM Computing Surveys*, 56(4):1–62, 2023.

[23] Robert Zubek. *Elements of Game Design*. The MIT Press, Aug 2020.